

Roll No.....

END SEMESTER EXAMINATION MAY-2025

Name of the Program: B. Tech (EE)

Semester: 8th

Name of the Course: Expert System & Fuzzy Logic

Course Code: TOE 801

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	What are the different stages involved in the development of an Expert System?	CO1
(b)	How do forward chaining and backward chaining work in Expert Systems? Explain with suitable diagrams.	
(c)	Discuss some limitations and challenges faced in Expert System development.	
Q2	(20 marks)	
(a)	Discuss the classifications of expert systems in artificial intelligence.	CO2
(b)	What are the practical uses of Expert Systems across different industries?	
(c)	How do computing techniques enhance Expert System functionalities?	
Q3	(20 marks)	
(a)	Compare Rule-Based Systems and Frame-Based Systems in knowledge representation. Explain with suitable examples.	CO3
(b)	Why are expert systems important at the present time? Explain with Examples and Figures.	
(c)	What are MYCIN and R1/XCON? How do they assist in building Expert Systems?	
Q4	(20 marks)	
(a)	Explain the inference engine structure of FLC with suitable examples and diagrams.	CO4
(b)	What are the Architecture and Operations of the FLC System? Explain with steps and block diagrams.	
(c)	What is the FUZZY LOGIC? And explain the properties of fuzzy sets with suitable equations.	

Q5	(20 marks)	CO4, CO5
(a)	Write a short note on the applications of the combination of Expert Systems and Fuzzy Logic.	
(b)	What is Defuzzification? Explain Defuzzification Methods with suitable examples and formulas.	
(c)	What are the applications of Expert Systems in AC and DC Drives? Explain some real-world use cases.	